Economics G6410 - Mathematical Methods for Economists - Fall 2018
Math Camp Exam

Department of Economics
Columbia University
Date: 6:10 - 8:00 p.m., September 4th, 2018
Instructions:
This is a closed-book, closed-note exam, and no calculator is allowed. There are 7 questions in total, and the full grade is 100 points.

1. (**10 pts**) Prove the following theorem.

Theorem (Mean Value Theorem)**.** *Let* $f : [a, b] \to \mathbb{R}$*, differentiable on* (a, b)*, and continuous on* $[a, b]$*. Then there exists* $x \in (a, b)$ *s.t.*

$$f'(x) = \frac{f(b) - f(a)}{b - a}$$

- **Solution**: First, let's consider the special case in which $f(a) = f(b)$, and we want to find $x \in (a, b)$ such that $f'(x) = 0$.
Because $[a, b]$ is compact and f is continuous on $[a, b]$, the function f has a maximum and a minimum. Also, there must exist $x \in (a, b)$ s.t. f achieves its maximum or minimum at x. I want to show that $f'(x) = 0$.
Suppose f achieves its maximum at $x \in (a, b)$. Arbitrarily take a sequence (x_n) convergent to x s.t. $x_n < x$ for any n. Then we have

$$f'(x) = \lim_{\tilde{x} \to x} \frac{f(\tilde{x}) - f(x)}{\tilde{x} - x} = \lim_{n \to \infty} \frac{f(x_n) - f(x)}{x_n - x} \geq 0$$

where the last inequality is because $f(x_n) - f(x) \leq 0$, $x_n - x < 0$, and $\leq$ is preserved in the limit.
Arbitrarily take a sequence (x'_n) convergent to x s.t. $x'_n > x$ for any n. Then we have

$$f'(x) = \lim_{\tilde{x} \to x} \frac{f(\tilde{x}) - f(x)}{\tilde{x} - x} = \lim_{n \to \infty} \frac{f(x'_n) - f(x)}{x'_n - x} \leq 0$$

Therefore, we must have $f'(x) = 0$ if f achieves its maximum at $x \in (a, b)$. Symmetrically, we can show that $f'(x) = 0$ if f achieves its minimum at $x \in (a, b)$. So we have proved the special case of the theorem where $f(a) = f(b)$.
If $f(a)$ and $f(b)$ are not necessarily equal, we can subtract the linear trend of f and define a new function $g : [a, b] \to \mathbb{R}$ as

$$g(x) := f(x) - \frac{f(b) - f(a)}{b - a} x$$

By definition, we have $g(a) = g(b)$. Applying the special case we have proved, we can find $x^* \in (a, b)$ s.t. $g'(x^*) = 0$. Because

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

for any $x \in (a, b)$, we have

$$f'(x^*) = \frac{f(b) - f(a)}{b - a}$$

$\square$

2. (**15 pts**) On a nonempty set X, define the discrete metric d, as

$$d(x,y) := \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}$$

(a) Verify that d as defined above is indeed a valid metric.

(b) Show that any subset of X is both open and closed.

(c) Show that a set S in X is compact iff it is finite.

- **Solution**:

(a) By definition of d, we have $d(x,y) = 0$ iff $x = y$, and also $d(x,y) = d(y,x)$ for any $(x,y) \in X^2$. To check the triangle inequality, take any $x, y, z \in X$. If $x = y$, $d(x,y) = 0 \leq d(x,z) + d(z,y)$. If $x \neq y$, $d(x,y) \leq 1 \leq d(x,z) + d(z,y)$. In either case, the triangle inequality holds.

(b) Take any $S \subset X$. WTS: (i) S is open, and (ii) S is closed.

i. Take any $x \in S$. Because $B_1(x) = \{x\} \subset S$, we know that x is an interior point of S. Therefore, S is open.

ii. It is sufficient to show that $S' = \emptyset$. Take any $x \in X$. WTS: x is not a limit point of S. Because $B_1(x) = \{x\}$, we have

$$(B_1(x) \setminus \{x\}) \cap S = \emptyset \cap S = \emptyset$$

and therefore x is not a limit point of S.

(c) ($\Rightarrow$) For any compact set S in X, consider the family of open balls $\{B_1(x)\}_{x \in S}$. Clearly, this is an open cover of S, since any $x \in S$ is covered by $B_1(x)$. By compactness of S, $\exists$ finite set $\hat{S} \subset S$ s.t. $\{B_1(x)\}_{x \in \hat{S}}$ is also an open cover of S, i.e., $\bigcup_{x \in \hat{S}} B_1(x) \supset S$. However, because

$$\bigcup_{x \in \hat{S}} B_1(x) = \bigcup_{x \in \hat{S}} \{x\} = \hat{S}$$

we have $\hat{S} \supset S$. Since $\hat{S} \subset S$ by construction of $\hat{S}$, we have $S = \hat{S}$, and is a finite set.

($\Leftarrow$) This direction is in fact true for general metric spaces. For any finite set S in (X, d), take any open cover $\{E_\alpha\}_{\alpha \in A}$ of S. For each $x \in S$, we have $x \in \bigcup_{\alpha \in A} E_\alpha$ by definition of open cover. So $\exists \alpha_x \in A$ s.t. $x \in E_{\alpha_x}$. Then $\{E_{\alpha_x}\}_{x \in S}$ also covers S, because for any $x \in S$ we have $x \in E_{\alpha_x}$ be construction of E_{α_x}. Because S is finite, $\{E_{\alpha_x}\}_{x \in S}$ is a finite subcover of S.

$\square$

3. (**10 pts**) In metric space (X, d), let (x_n) be a Cauchy sequence. Prove that (x_n) is a bounded sequence.

- **Solution**: Let (x_n) be a Cauchy sequence. Because (x_n) is Cauchy, there exists $N \in \mathbb{N}$ s.t. $d(x_m, x_n) < 1$ for any $m, n > N$. Therefore $x_n \in B_1(x_{N+1})$ for any $n > N$. Let

$$r := \max\{d(x_1, x_{N+1}), d(x_2, x_{N+1}), ..., d(x_N, x_{N+1})\} + 1$$

Clearly we have $x_n \in B_r(x_{N+1})$ for any $n \in \mathbb{N}$, and so the sequence (x_n) is bounded. $\square$

4. (**20 pts**) Diagonalization

   (a) Is the matrix
$$A = \begin{bmatrix} 2 & -1.5 \\ 1 & -0.5 \end{bmatrix}$$
   diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If so, diagonalize it, i.e., find a 2×2 invertible matrix P s.t. $P^{-1}AP = \Lambda$, where Λ is a diagonal matrix.
   Consider a 2×2 real matrix as a vector in $\mathbb{R}^4$, and use d_2 on $\mathbb{R}^4$ to measure distance between two 2×2 real matrices. Find $\lim_{n \to \infty} A^n$.

   (b) Is the matrix
$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$
   diagonalizable in $\mathbb{C}$? If so, diagonalize it.

- **Solution**:

  (a) The characteristic polynomial of A is
$$\begin{aligned} \det(\lambda I - A) &= \det\left(\begin{bmatrix} \lambda - 2 & 1.5 \\ -1 & \lambda + 0.5 \end{bmatrix}\right) \\ &= (\lambda - 2)(\lambda + 0.5) - 1.5(-1) \\ &= (\lambda - 1)(\lambda - 0.5) \end{aligned}$$
  Therefore the two eigenvalues of A is $\lambda_1 = 1$ and $\lambda_2 = 0.5$.
  By definition, $x_1 \in \mathbb{C}^2 \setminus \{0\}$ is an eigenvector that corresponds to $\lambda_1 = 1$ iff $Ax_1 = \lambda_1 x_1$, i.e.,
$$(\lambda_1 I - A)\, x_1 = 0$$
  i.e.,
$$\begin{bmatrix} -1 & 1.5 \\ -1 & 1.5 \end{bmatrix} x_1 = 0$$
  So an eigenvector that corresponds to $\lambda_1 = 1$ is $x_1 = (3, 2)$.
  Similarly, we can find an eigenvector that corresponds to $\lambda_2 = 0.5$ as $x_2 = (1, 1)$.
  Therefore, we have
$$\begin{aligned} A \begin{bmatrix} x_1 & x_2 \end{bmatrix} &= \begin{bmatrix} Ax_1 & Ax_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \end{aligned}$$
  Define
$$P := \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

and

$$\Lambda := \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$$

and so we have $AP = P\Lambda$.

Because the two eigenvectors $x_1 = (3, 2)$ and $x_2 = (1, 1)$ are linearly independent, the matrix P is invertible. Therefore, we have

$$P^{-1}AP = P^{-1}(P\Lambda) = \left(P^{-1}P\right)\Lambda = \Lambda$$

Because P and Λ are real matrices, A is diagonalizable in both $\mathbb{C}$ and $\mathbb{R}$.

Also, we have

$$A = A\left(PP^{-1}\right) = (AP)P^{-1} = P\Lambda P^{-1}$$

and so

$$A^n = \left(P\Lambda P^{-1}\right)^n = P\Lambda^n P^{-1}$$

Because A^n is a linear function of Λ^n (each entry of 4 entries of A^n is a linear combination of the 4 entries of Λ^n), and so A^n is continuous in Λ^n. Therefore, we have

$$\begin{aligned}
\lim_{n\to\infty} A^n &= P\left(\lim_{n\to\infty}\Lambda^n\right)P^{-1} = P\left(\lim_{n\to\infty}\begin{bmatrix} 1^n & 0 \\ 0 & 0.5^n \end{bmatrix}\right)P^{-1} \\
&= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} \lim_{n\to\infty} 1^n & 0 \\ 0 & \lim_{n\to\infty} 0.5^n \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} 3 & 0 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} \\
&= \begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix}
\end{aligned}$$

(b) The characteristic polynomial of A is

$$\begin{aligned}
\det(\lambda I - A) &= \det\left(\begin{bmatrix} \lambda & -1 \\ 1 & \lambda \end{bmatrix}\right) \\
&= \lambda^2 - 1(-1) \\
&= (\lambda - i)(\lambda + i)
\end{aligned}$$

Therefore the two eigenvalues of A is $\lambda_1 = i$ and $\lambda_2 = -i$. Since A has distinct eigenvalues, the matrix A is diagonalizable in $\mathbb{C}$.

By definition, $x_1 \in \mathbb{C}^2 \setminus \{0\}$ is an eigenvector that corresponds to $\lambda_1 = i$ iff $Ax_1 = \lambda_1 x_1$, i.e.,

$$(\lambda_1 I - A)x_1 = 0$$

i.e.,

$$\begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix} x_1 = 0$$

So an eigenvector that corresponds to $\lambda_1 = 1$ is $x_1 = (i, -1)$.
Similarly, we can find an eigenvector that corresponds to $\lambda_2 = -i$ as $x_2 = (i, 1)$.
Therefore, we have

$$\begin{aligned} A \begin{bmatrix} x_1 & x_2 \end{bmatrix} &= \begin{bmatrix} Ax_1 & Ax_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \end{aligned}$$

Define

$$P := \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} i & i \\ -1 & 1 \end{bmatrix}$$

and

$$\Lambda := \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$$

and so we have $AP = P\Lambda$.
The two eigenvectors $x_1 = (i, -1)$ and $x_2 = (i, 1)$ are linearly independent. To see this, suppose we have $\lambda_1, \lambda_2 \in \mathbb{C}$ s.t. $\lambda_1 x_1 + \lambda_2 x_2 = 0$, i.e.,

$$((\lambda_1 + \lambda_2)\, i, -\lambda_1 + \lambda_2) = (0, 0)$$

Then we have $\lambda_1 = \lambda_2$, and so $2\lambda_1 i = 0$. Because $i \neq 0$, we have $\lambda_1 = \lambda_2 = 0$.
Therefore, the matrix P is invertible, and we have

$$P^{-1}AP = P^{-1}(P\Lambda) = \left(P^{-1}P\right)\Lambda = \Lambda$$

□

5. (**15 pts**) Log-linearization

   (a) Verify that the log-linearization of $Y = AB$ is $\hat{Y} = \hat{A} + \hat{B}$ (using First Order Taylor Approximations).

   (b) Log-linearize $Y = A^B$, where the two variables A and B only take positive values.

   (c) Log-linearize $Y = A^\alpha + B^\beta$, where α and β are constants. You may use the shortcuts.

   - **Solution**:
     (a) Taking logs we get :

     $$\log Y = \log A + \log B$$

     Taking a first order Taylor expansion of each term and noting that $\log Y^* = \log A^* + \log B^*$ we get :

$$\begin{aligned}\log Y^* + \frac{Y - Y^*}{Y^*} &= \log Y^* + \frac{A - A^*}{A^*} + \log B^* + \frac{B - B^*}{B^*} \\ \implies \hat{Y} &= \hat{A} + \hat{B}\end{aligned}$$

where $\hat{x} = \frac{x - x^*}{x^*}$

(b) Taking logs, we get :

$$\log Y = B \log A$$

Taking the First Order Taylor approximation, we get :

$$\begin{aligned}\log Y^* + \frac{Y - Y^*}{Y^*} &= B^* \log A^* + B^* \log A^* \frac{(B - B^*)}{B^*} + \frac{B^*}{A^*}(A - A^*) \\ \implies \hat{Y} &= B^* \log A^* \hat{B} + B^* \hat{A} \\ &= Y^* \hat{B} + B^* \hat{A}\end{aligned}$$

(c)

$$\begin{aligned}\hat{Y} &= \frac{A^{*\alpha}}{A^{*\alpha} + B^{*\alpha}} \hat{A^{\alpha}} + \frac{B^{*\alpha}}{A^{*\alpha} + B^{*\alpha}} \hat{B^{\alpha}} \\ \implies \hat{Y} &= \frac{\alpha A^{*\alpha}}{A^{*\alpha} + B^{*\alpha}} \hat{A} + \frac{\alpha B^{*\alpha}}{A^{*\alpha} + B^{*\alpha}} \hat{B}\end{aligned}$$

6. (**10 pts**) Let $x \in \mathbb{R}^n$ and $r > 0$. Show that the open ball $B_r(x)$ in $(\mathbb{R}^n, d_2)$ is a convex set in the vector space $\mathbb{R}^n$. (Hint: triangle inequality of norm)

- **Solution**: Take any $x_1, x_2 \in B_r(x)$ and $\lambda \in [0,1]$. WTS: $\lambda x_1 + (1-\lambda) x_2 \in B_r(x)$.

$$\begin{aligned}d_2(\lambda x_1 + (1-\lambda) x_2, x) &= \|\lambda x_1 + (1-\lambda) x_2 - x\| \\ &= \|\lambda (x_1 - x) + (1-\lambda)(x_2 - x)\| \\ &\leq \|\lambda (x_1 - x)\| + \|(1-\lambda)(x_2 - x)\| \\ &= \lambda \|x_1 - x\| + (1-\lambda)\|x_2 - x\|\end{aligned}$$

If $\lambda = 0$, we have $\lambda\|x_1 - x\| + (1-\lambda)\|x_2 - x\| = \|x_2 - x\| < r$.
If $\lambda = 1$, we have $\lambda\|x_1 - x\| + (1-\lambda)\|x_2 - x\| = \|x_1 - x\| < r$.
If $\lambda \in (0,1)$, we have

$$\lambda\|x_1 - x\| + (1-\lambda)\|x_2 - x\| < \lambda r + (1-\lambda) r = r$$

Therefore, we always have $d_2(\lambda x_1 + (1-\lambda) x_2, x) < r$, i.e., $\lambda x_1 + (1-\lambda) x_2 \in B_r(x)$.

7. (**20 pts**) Consider the following problem:

$$\max_{(x_1, x_2) \in \mathbb{R}_+^2} x_1^{\alpha} x_2^{\beta}$$

s.t.

$$p_1 x_1 + p_2 x_2 \leq m$$

where $\alpha, \beta \in (0,1)$, $r \in (0,1)$, $p \in \mathbb{R}^2_{++}$ and $m \in \mathbb{R}_{++}$ are parameters.

(a) Does a solution exist?

(b) Prove that the budget constraint is binding at any possible maximum.

(c) Argue that the constraints $x_1 \geq 0$, $x_2 \geq 0$ are not binding at any possible maximum.

(d) Solve the program.

- **Solution**:

  (a) Yes a solution exists by Weierstrass' Theorem since the objective function is continuous on its domain and the constraint set :

  $$D = \left\{(x_1, x_2) \in \mathbb{R}^2_+ | p_1 x_1 + p_2 x_2 \leq m\right\}$$

  is a compact and non-empty set. Hence a solution exists by W-T.

  (b) Suppose by contradiction that the constraint does not bind at the optimum say (x_1^*, x_2^*) i.e. we have

  $$p_1 x_1^* + p_2 x_2^* < m$$

  Set $\varepsilon = \frac{m - p_1 x_1 - p_2 x_2}{p_1}$. Then we must have :

  $$p_1 \left(x_1^* + \varepsilon\right) + p_2 x_2^* = m$$

  i.e. the bundle $(x_1^* + \varepsilon, x_2^*)$ is a feasible bundle. Since the objective function is strictly increasing in both its arguments (barring at the axes which we can rule out since $f(\varepsilon, \varepsilon) > f(a, 0) = f(0, a)$ for any $a \in \mathbb{R}_+$ for arbitrarily small ε), we have that :

  $$\left(x_1^* + \varepsilon\right)^\alpha x_2^{*\beta} > x_1^{*\alpha} x_2^{*\beta}$$

  which contradicts that (x_1^*, x_2^*) is a maximum.

  (c) As argued above, we can rule out any solutions of the form $(a, 0)$ or $(0, a)$ since $f(\varepsilon, \varepsilon) > f(a, 0) = f(0, a)$ for any $a \in \mathbb{R}_+$ for arbitrarily small ε.

  (d) Since we proved that $x_i > 0$, the Lagrangean is :

  $$L\left(x_1, x_2, \lambda\right) = x_1^\alpha x_2^\beta + \lambda \left(m - p_1 x_1 - p_2 x_2\right)$$

  Differentiating wrt (x_1, x_2, λ) and solving the FOC's we get :

  $$x_1^* = \frac{\alpha m}{(\alpha + \beta) p_1}$$
  $$x_2^* = \frac{\beta m}{(\alpha + \beta) p_2}$$